Review: Data types and variability

Presentation Outline

Data Table (a.k.a., Data Set)

Observations: Represent the rows of a data table. In specific contexts, these may also be called *cases*, *records*, *subjects*, or *participants*.

Alternatively, an **observation** refers to the value of a variable for a specific individual.

Variables: Any characteristic of an individual that we are interested in. A variable typically takes on different values for different individuals.

We distinguish between:

- Categorical (qualitative) variables
- Numerical (quantitative) variables

	A	B	C	D	E	F	G	H
1	Patient ID	Gender	Treatment Group	Age	Blood Pressure (mmHg)	Cholesterol (mg/dL)	Symptom Severity	Outcome
2	P001	Female	Control	34	120	190	Mild	Improved
3	P002	Male	Treatment A	45	138	220	Moderate	Stable
4	P003	Female	Treatment B	29	115	180	Mild	Improved
5	P004	Male	Control	54	142	245	Severe	Worsened
6	P005	Female	Treatment A	41	130	200	Moderate	Improved
7	P006	Male	Treatment B	38	128	210	Mild	Stable
8	P007	Female	Control	60	145	230	Severe	Worsened
9	P008	Male	Treatment A	50	135	215	Moderate	Stable
10	P009	Male	Treatment A	48	138	226	Moderate	Stable
11	P010	Female	Control	48	129	214	Mild	Worsened
12	P011	Male	Treatment A	68	139	194	Moderate	Improved
13	P012	Female	Treatment A	56	138	206	Moderate	Stable
14	P013	Female	Control	25	126	175	Severe	Worsened
15	P014	Female	Treatment B	45	141	233	Severe	Improved
16	P015	Male	Treatment A	66	140	222	Moderate	Stable
17	P016	Female	Treatment A	35	140	240	Severe	Stable
18	P017	Male	Treatment B	38	133	249	Moderate	Improved
19	P018	Male	Treatment A	49	150	189	Moderate	Improved
20	P019	Male	Control	36	117	209	Moderate	Worsened
21	P020	Male	Treatment B	49	134	229	Moderate	Improved
22	P021	Female	Treatment A	46	137	180	Moderate	Improved

Categorical Variables

A categorical variable identifies group membership.

More formally, it is a variable that can take on one of a limited and usually fixed number of possible values, assigning each individual to a particular group based on a qualitative property.

Example

ID	Diagnosis	Weight (kg)
001	Asthma	70
002	Diabetes	83
003	None	65

Two types:

- **Nominal**: no inherent order
- **Ordinal**: categories have a meaningful order

Nominal Variables

A nominal variable has values with no natural (meaningful) ordering.

- Medical diagnosis (diabetes, hypertension, asthma, none)
- Type of stock (small cap, large cap, etc.)
- Operating system: Windows, macOS, Linux
- Phone numbers, social security numbers (used for identification, not computation)
- Ecosystem type: forest, grassland, desert, wetland
- Team name: Yankees, Lakers, Packers

Ordinal Variables

An ordinal variable has values that can be meaningfully ordered.

- Pain severity: none, mild, moderate, severe
- Grades: A, A-, B+, B, B-, ...
- Education level: no diploma, high school, college degree, graduate degree
- Customer loyalty: bronze, silver, gold, platinum
- Air quality: good, moderate, unhealthy, very unhealthy, hazardous
- Division level: junior, varsity, elite

Example: Likert Scale

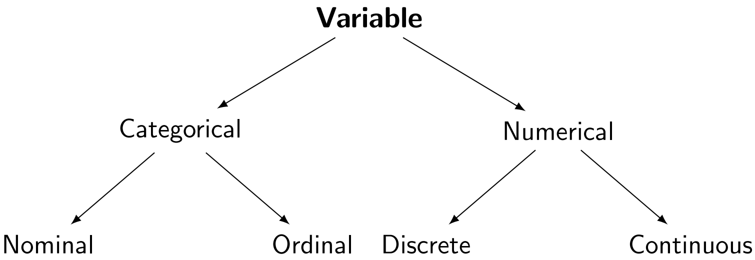
Strongly Disagree			Neutral			Strongly Agree
1	2	3	4	5	6	7

Numerical Variables

A numerical (quantitative) variable takes values for which arithmetic operations (e.g., adding, averaging) are meaningful.

- Height/weight of a person
- Temperature
- Time to run a mile
- Currency exchange rates
- Number of webpage hits in an hour

Visual Summary: Types of Variables



Key Distinction

Categorical: describes qualities or groups

Numerical: expresses quantities or measurements

What is Variation?

Definition

Variation refers to differences in a characteristic among individuals or items. It may also describe fluctuation over time or space.

- Stock values vary daily
- Sales fluctuate across weeks
- Blood pressure varies across time and individuals
- Customer preferences differ
- Response times vary with server load and location
- A player's performance varies across games

Some First Facts about Variation

- Individuals differ in physical and behavioral traits
- Repeated measures on the same person also vary
- Variability arises from many causes (natural, measurement error, etc.)
- Both qualitative and quantitative variables exhibit variation
- Some things vary a little; some, a lot

Variability is what makes decisions in the face of uncertainty so difficult. It is also what makes statistics interesting and powerful.

— *Gould (2004)*

Distribution of a Variable

Definition

The distribution of a variable describes the values it can take and how often each occurs.

Common Tool

A **frequency table** summarizes how often each value or category appears in the data.

Different types of variables require different visual and numerical tools for analysis.

Example: Frequency Table (Categorical)

Favorite Fruit Survey (n = 100)

Fruit	Frequency
Apple	30
Banana	25
Orange	20
Grapes	15
Other	10

Summary: Key Concepts

- **Observations** are the rows in a data table; each represents a unit and its measured variables.
- **Variables** can be:
  - **Categorical:** Nominal (no order) or Ordinal (ordered)
  - **Numerical:** values where calculations make sense
- **Variation** is everywhere — across individuals, across time, across space...
- **Distribution** tells us what values a variable takes and how often
- Understanding variable type helps determine which tools and analyses are appropriate